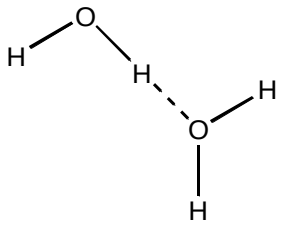
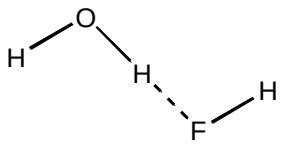
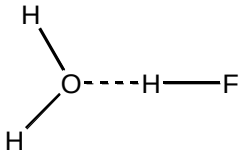
Chem 150 Answers Problem Set 3 Intermolecular Interactions and Colligative Properties

1. To answer the questions below use solid lines as usual for covalent bonds and dotted lines for any hydrogen bonds. Be sure to align the molecules properly. Use perspective drawings clearly showing the shape of the participating molecules and the way they interact (align!).
   a) Draw two water molecules that are hydrogen bonded.

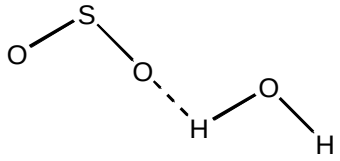


   b) Draw a water molecule that is hydrogen bonded to a hydrogen fluoride molecule. Is there more than one possible interaction? If so give a perspective drawing.





   c) Is there any hydrogen bonding between sulfur dioxide and water? If so, give a perspective drawing. If not, state why not.

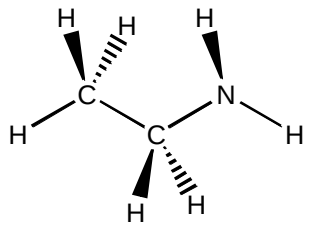


This drawing should be corrected to account for a 180 ° hydrogen bond angle.

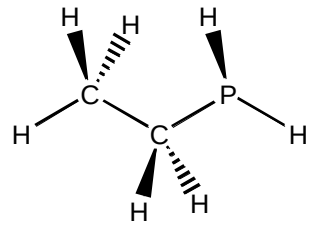
   d) Is there any hydrogen bonding between methane and water? If so, give a perspective drawing. If not, state why not.

   No. The hydrogens in methane are not bound to a highly electronegative element so they cannot participate in hydrogen bonding

2. Shown below are the perspective drawings of two relatively simple organic compounds A and B. Both compounds have the same molecular shape differing only in the identity of one atom. Identify the high and low boiling compound. In one or two sentences give the reason for your choice. Use arguments involving intermolecular interactions.



Compound A



Compound B

Compound A is the higher boiling compound. Even though its mass is slightly lower than that of compound B it has the higher boiling point because it contains a hydrogen bound to nitrogen which can engage in hydrogen bonding.

3. Arrange in order of increasing boiling point: F_2, Cl_2, SO_2, He, N_2.

He, N_2, F_2, Cl_2, SO_2 (SO_2 is polar! All other species are non-polar)

4. Calculate the osmotic pressure of a solution made by dissolving 50 g glucose ($C_6H_{12}O_6$) in enough water to make 2.0 L of solution. Assume 25 °C. R = 8.314 L kPa mol^{-1} K^{-1}

First we find the molecular mass of glucose: 180 g/mol so we can convert grams to moles:

50 g : 180g/mol = 0.278 mol M = n/V = 0.278 mol/2.0 L = 0.139 mol/L

Π = MRT = (0.139 mol/L) x (8.314 L kPa/mol K) x (298 K)

Π = 344 kPa

5. Calculate the vapour pressure of a solution made by mixing 1 mol of the non volatile glycerine (propanetriol) with 10 mol of water at 25 °C. The vapour pressure of pure water at that temperature is 23.8 torr.

This is a solution of a non volatile liquid in water. The vapour pressure of the solution is given by Raoult's law (where x_{water} is the mol fraction of water in the solution).

$P_{solution} = x_{water}P°_{water}$ $x_{water} = \frac{n(water)}{n(water) + n(glycerine)} = \frac{10\ mol}{10\ mol + 1\ mol} = 0.909$

$P_{solution}$ = 0.909 x 23.8 torr = 21.6 torr

6. The vapour pressures of pure toluene (C_7H_8) and pure benzene (C_6H_6) at 25 °C are 28.4 and 95.1 torr respectively. A solution is prepared in which 100 g of benzene are mixed with 100 g of toluene.
   a) What are the partial pressures of benzene and toluene above the solution?
   b) What is the total vapour pressure?
   c) What is the composition of the vapour above the mixture at that temperature?

   to a) Before we can apply Raoult's law we need to find the mole fractions of each component in the mixture.

$$x_{tol} = \frac{n(tol)}{n(tol)+n(benz)} \quad \text{with} \quad n_{tol} = 100 \text{ g toluene x } \frac{mol}{92.14\ g} = 1.085 \text{ mol}$$

$$\text{and} \quad n_{benz} = 100 \text{ g benzene x } \frac{mol}{78.11\ g} = 1.280 \text{ mol}$$

$$x_{tol} = \frac{1.085\ mol}{1.085\ mol+1.280\ mol} = 0.4588$$

and

$$x_{benz} = \frac{n(benz)}{n(benz)+n(tol)} = \frac{1.280\ mol}{1.280\ mol+1.085\ mol} = 0.5412$$

   Check: the mol fractions of all components of a mixture must add up to 1 (0.4588 + 0.5412 = 1)
   Now we can calculate the individual vapour pressures of each component:

   $P_{benz} = x_{benz}P°_{benz}$ = 0.5412 x 95.1 torr = 51.47 torr

   $P_{tol} = x_{tol}P°_{tol}$ = 0.4588 x 28.4 torr = 13.03 torr

   to b) Using Daltons Law of partial pressures gives the total vapour pressure over the mixture:

   $P_{total} = P_{benz} + P_{tol}$ = 64.50 torr

   to c) The composition of the vapour is $x_{benzene}$ (vapour) = $\frac{p(benzene)}{P(total)} = \frac{51.47\ torr}{64.5\ torr} = 0.798$

   and $x_{toluene}$(vapour) = 1-0.798 = 0.202

   As expected the vapour is enriched in the more volatile benzene.

7. Nitrogen gas (N_2) dissolves in water only slightly (e.g. a few ppm at room temperature) whereas gaseous ammonia (NH_3) dissolves very well in water (e.g. several hundred grams per litre). In a short statement give the reason for the different solubilities.

   Ammonia, like water, is highly polar whilst dinitrogen is a nonpolar hydrophobic gas. There are many hydrogen bond possible between ammonia and water but not between dinitrogen and water.

8. Under an O_2(g) pressure of 1 atm, 28.31 mL of O_2(g) dissolves in 1.00 L H_2O at 25 °C. What will be the molarity of O_2 in a saturated solution at 25 °C when the O_2(g) pressure is increased to 3.86 atm. Assume the solution volume remains at 1.00 L.

Using the ideal gas equation we find the Molarity of the saturated oxygen solution at 1 atm and 25°C:

n = $\frac{pV}{RT}$ = $\frac{(101.3\ kPa)(0.02831\ L)}{(8.314\ L\ kPa\ mol^{-1}K^{-1})(298\ K)}$ = 0.00116 mol

M = $\frac{n}{V}$ = $\frac{0.00116\ mol}{L}$ = 0.00116 mol/L

This is a Henry's Law problem. We use the data in the first part of the question to calculate the Henry's law constant for oxygen at the given temperature in water and use the constant *k* to find the solubility of oxygen at the higher pressure.

$S_g = k\ p_g$ k = $\frac{0.00116\ \mathrm{mol/L}}{1\ atm}$ = 0.00116 mol L^{-1} atm^{-1}

S_g(@3.86 atm) = $k\ p_g$ = (0.00116 mol L^{-1} atm^{-1}) (3.86 atm) = 4.47 x 10^{-3} mol/L

9. Using data from the example above, determine the molarity of O_2 in an aqueous solution at equilibrium with air at normal atmospheric pressure. The volume percent of oxygen in air 20.95 % vol.

   First we find the partial pressure of oxygen: p_{oxygen} = $\frac{20.95\ mL}{100\ mL}$ p_{total} = 0.2095 atm

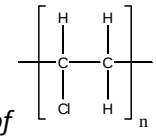
   Next we use Henry's law: $S_g = k\ p_g$ = (0.00116 mol L^{-1} atm^{-1}) (0.2095 atm) = 2.43x10^{-4} mol/L

10. A 0.72 g sample of polyvinyl chloride (PVC) is dissolved in 250.0 mL of a suitable solvent at 25 °C. The solution has an osmotic pressure of 1.67 mm Hg. What is the molar mass of the PVC?

    Since molarity M is related to numbers of moles (per volume of solution) and the number of moles is related to the molar mass (n = $\frac{m}{Mw}$) we can extract molar masses (molecular weights, MW) from osmotic pressure measurements.

    π = MRT = $\frac{nRT}{V}$ = $\frac{mRT}{MwV}$ solving for the molecular weight and converting pressure from mm Hg to kPa gives: 1.67 mm Hg $\frac{101.3\ kPa}{760\ mm\ Hg}$ = 0.223 kPa

    Mw = $\frac{mRT}{\pi V}$ = $\frac{(0.00072\ kg)(8.315\ L\ kPa\ mol^{-1}K^{-1})(298\ K)}{(0.223\ kPa)(0.250\ L)}$ = 32 kg/mol = 32000 g/mol

    Polyvinylchloride is made up of repeating units of 

    Each monomer unit, made up of the same C_2H_3Cl group, has a mass of

(2x 12.01 + 3 x 1.008 + 35.45) g/mol = 62.47 g/mol

"n" the number of repeating units in the polymer formula above must thus be about $\frac{32\,000}{62.47}$ *= 512 which is a low to moderate molecular weight for a polymer.*

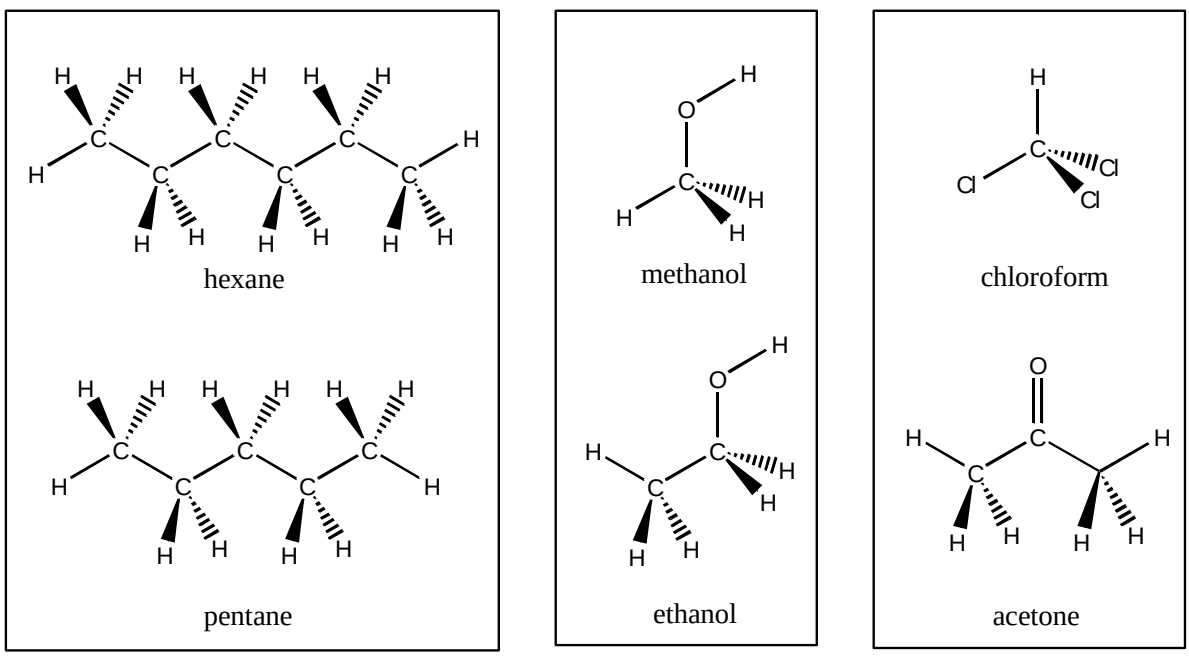
11. The density of acetone vapour in equilibrium with liquid acetone, $(CH_3)_2CO$, at 32 °C is 0.876 g/L. What is the vapour pressure of acetone at 32 °C, expressed in kilopascals

    First we find the number of moles of acetone in one Litre of vapour and then we use the ideal gas equation to calculate the vapour pressure.

    $n_{acetone}$ (in 1L vapour) = $\frac{0.876\ g}{58.08 g\ mol^{-1}}$ = 0.01508 mol

    p = $\frac{nRT}{V}$ = $\frac{(0.01508\ \text{mol})(8.314\ L\ kPa\ mol^{-1}K^{-1})(305\ K)}{(1L)}$ = 38.24 kPa

12. Solutions can be prepared from the following pairs of solvents: Hexane and pentane (D), Ethanol and Methanol (E) and acetone chloroform (F)



D E F

a) (**2 marks**) All of the above solvent pairs are <u>completely miscible</u>. Explain why.

Likes dissolves likes! In D we have two nonpolar hydrocarbons of similar size. The London dispersion forces between molecules of C_6H_{14} will be similar to those between molecules of C_5H_{12} and most importantly for miscibility: they will also be of similar strength between molecules of C_6H_{14} and C_5H_{12}.

In E and F we have pairs of molecules that are both polar. In E the molecules are capable of hydrogen bonding and in F dipol-dipol interactions are possible. In each case (pair) the intermolecular interactions between the same molecules are similar in strength when compared to the intermolecular interactions between different molecules in the mixture. Thus D, E and F are completely miscible.

b) (**3 marks**) When preparing the actual solutions it is observed that upon mixing chloroform and acetone (F) considerable amounts of heat are generated that is, the solution is warmer then the two components before mixing. No such heat is produced by mixing either of the other systems (hexane-pentane and methanol-ethanol). Account for this observation involving intermolecular interactions.

<u>This was a hard question!</u> Intermolecular interactions between C_6H_{14} and C_5H_{12} are **<u>very</u>** similar to those between C_6H_{14} and C_6H_{14} and between C_5H_{12} and C_5H_{12} so in the mixing process the energy required to separate (dilution!) molecules of C_5H_{12} and C_5H_{12} (see figure M) as well as C_6H_{14} and C_6H_{14} (see figure N) will be almost identical to the energy gained by bringing molecules of C_6H_{14} and C_5H_{12} back together (FigureQ) and no heat is gained or lost by the mixture. The same holds for mixture E.

For **F** the situation is somewhat different. The mixture is heating up can only mean that the interactions between molecules of acetone and chloroform are much stronger (see (2) Figure R) than those between chloroform and chloroform (3) and those between acetone and acetone (1). The energy is released as heat to the surroundings and the solution warms upon mixing.

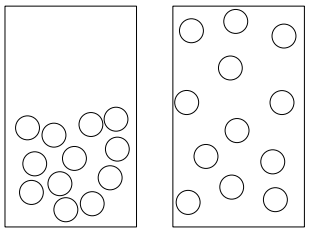


Figure M: energy is required to separate solvent molecules

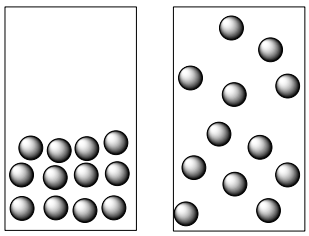


Figure N

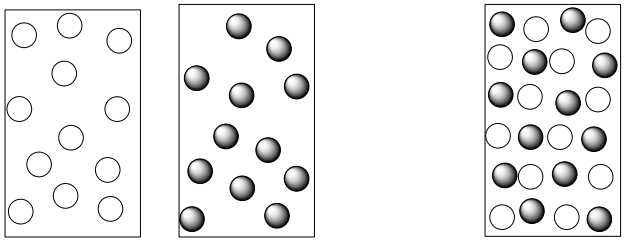


Figure Q: energy (heat) is released by bringing the molecules back to close contact (intermolecular forces are much stronger at close proximity and quickly decrease as separation increases which is why we assume no intermolecular interaction in an ideal gas where the separation between molecules is much larger then in liquids or solids.)

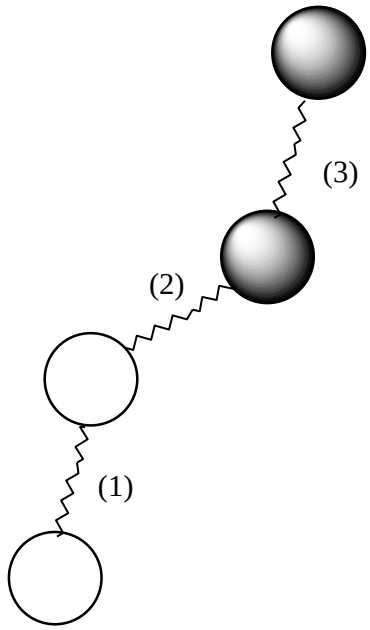


Figure R Intermolecular interactions between the components of the mixture F: (1) acetone-acetone, (3) chloroform-chloroform, (2) acetone-chloroform

Formulas and constants:

$pV = nRT$

$p_iV_i(T_i)^{-1} = p_fV_f(T_f)^{-1}$

$P_{total} = p_1 + p_2 + p_3 + ... + p_n$

$[p + n^2a/V^2][V - nb] = nRT$

$V_{mol} = 22.4$ L mol^{-1} at STP (1 atm = 101.325 kPa = 760 mm Hg and 273.15 K)

$R = 8.3145$ L kPa mol^{-1} K^{-1}

Avogadro's number $N_a = 6.022 \times 10^{23}$ mol^{-1}

Unified atomic mass unit 1 u = 1.660×10^{-27} kg

$S_g = kp_g$

$\Pi = MRT$

$P_A = x_AP°_A$